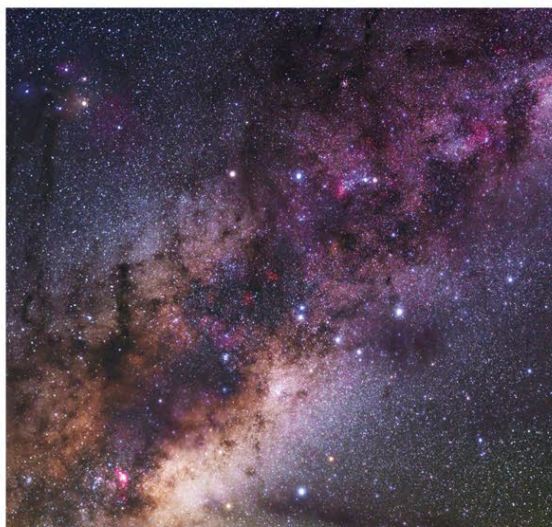


CELESTIAL OBJECTS

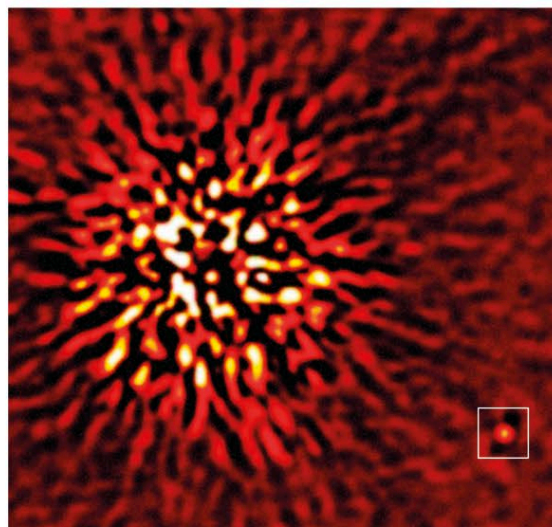
SCORES OF DIFFERENT TYPES OF OBJECTS EXIST OUT IN SPACE, RANGING FROM COSMIC RAYS—CHARGED SUBATOMIC PARTICLES WHIZZING AROUND AT EXTREME SPEED—TO VAST, MAJESTIC GALAXY CLUSTERS.

Stars are by far the most numerous objects that can actually be seen, because they emit their own light. Most other observable features of the night sky either consist mainly of stars (galaxies and star clusters) or are visible because they reflect starlight (planets, moons, and comets, for example). In addition, various extremely dim or entirely dark objects, such as brown dwarfs and black holes, are out there, but vary from extremely hard to near-impossible to detect.



△ Stars

A star is an extremely hot ball of gas that generates energy through nuclear fusion of hydrogen (and sometimes other elements). All nearby stars are part of the Milky Way Galaxy, which (as shown above) appears as a band across the night sky.



△ Brown dwarfs

Brown dwarfs are “nearly-stars.” They are more massive than most planets, but not massive enough to sustain the nuclear fusion of ordinary hydrogen, as stars do. This image reveals the dim glow from a brown dwarf (boxed) orbiting a Sun-like star.



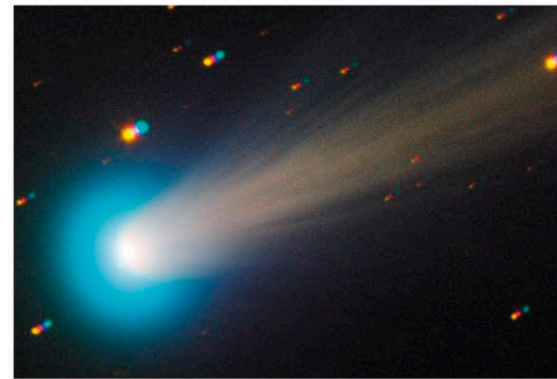
△ Star clusters

A star cluster is a large collection of stars bound together by gravity. Several thousand have been identified in our galaxy, and they fall into two types: globular clusters (like the one shown above) and open clusters.



△ Star remnants

When giant stars die, they leave various types of remnant. This always includes a compact remnant of the original star's core. However, this ghostly looking object is gas and dust debris ejected from a star when it exploded as a supernova.



△ Comets

Comets are chunks of ice and rock that orbit in the far reaches of the Solar System. A few stray close to the Sun—some at regular intervals. Frozen chemicals in the comet then vaporize to produce a glowing coma (head) and long dust and gas tails.

Nebulae

Nebulae are clouds of gas and dust in the vast expanses of space between stars. Many contain regions of star formation. In some, light from hot newborn stars excites gas atoms in the nebula, which then begin to emit light in various colors. An example of one of these colourful objects is the Carina Nebula, shown here. It is a prominent naked-eye sky feature in the Southern Hemisphere.

